

Secondary Module: Synthetic Ultimate Consumer (Silpo+Novus+ConsumerUA)

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely I(1)-like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

Built after primary chain models.

Coverage is restricted to each standardized_type ProZorro period.

frequency=daily

series_rows=1890

model_rows=10

Table: SyntheticConsumer_Models

standardized_type	frequency	model_family	link	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	model_status	notes
cottage_cheese	daily	ARDL	prozorro_to_ultimate_cons...	ultimate_consumer_price	prozorro	42	-0.4066263125016476	-1.685428485355531	nan	nan	nan	nan	ok	bounds_lower_p=0.02475; b...
cream	daily	ARDL	producer_to_prozorro	prozorro	producer	199	3.349348015760423	0.266717232238544	nan	nan	nan	nan	unreliable	bounds_lower_p=0; bounds_...
cream	daily	ARDL	producer_to_ultimate_cons...	ultimate_consumer_price	producer	46	-10.082147233462337	-7.788589449718949	nan	nan	nan	nan	ok	bounds_lower_p=6.253e-05;...
cream	daily	ECM	producer_to_ultimate_cons...	ultimate_consumer_price	producer	43	-5.388499561865965	-1.449855506085529	-1.3992144686360841	0.00040790650958883097	nan	nan	ok	engle_granger_p=5.74e-05;...
hard_cheese	daily	ARDL	producer_to_ultimate_cons...	ultimate_consumer_price	producer	189	1.602343683487597	2.538978832015522	nan	nan	nan	nan	unreliable	bounds_lower_p=2.458e-05;...
milk	daily	ARDL	producer_to_prozorro	prozorro	producer	201	0.6534485521618496	-0.28996329386924236	nan	nan	nan	nan	ok	bounds_lower_p=0.005587; ...
milk	daily	ECM	producer_to_prozorro	prozorro	producer	197	-0.002155003969288638	-0.3006834081052232	-0.9538372381620318	3.2952288002155387e-21	nan	nan	unreliable	engle_granger_p=1.912e-24...
milk	daily	ARDL	producer_to_ultimate_cons...	ultimate_consumer_price	producer	201	1.0305554412349078	1.2107328897564298	nan	nan	nan	nan	unreliable	bounds_lower_p=0.3593; bo...
sour_cream	daily	ARDL	producer_to_prozorro	prozorro	producer	178	0.0982872243249222	0.36158068419167616	nan	nan	nan	nan	ok	bounds_lower_p=4.545e-05;...
sour_cream	daily	ARDL	producer_to_ultimate_cons...	ultimate_consumer_price	producer	178	1.005497411785936	1.0100286454509504	nan	nan	nan	nan	ok	bounds_lower_p=0.4192; bo...

Table: SyntheticConsumer_PreTests

standardized_type	series	frequency	adf_level_p	kpss_level_p	adf_diff1_p	kpss_diff1_p	adf_diff2_p	kpss_diff2_p	integration_class	stability_flag
butter	producer	daily	0.28733427647950704	0.1	3.939989249212331e-07	0.1	2.1006274517545675e-13	0.1	I(1)	0
	prozorro	daily	0.056404226942647726	0.01	3.518946761497874e-22	0.1	1.9269184182026386e-13	0.1	I(1)	1
butter	ultimate_consumer_price	daily	0.9948280909044506	0.01629787475740224	2.332614295314388e-05	0.1	0.0030906664317012153	0.1	I(1)	1
cottage_cheese	producer	daily	nan	nan	nan	nan	nan	nan	ambiguous	0
cottage_cheese	prozorro	daily	1.1483227108295172e-10	0.01	4.589982408975803e-17	0.1	4.813752592554151e-10	0.1	I(1)	0
cottage_cheese	ultimate_consumer_price	daily	0.8122416324953605	0.07937483460730256	3.111052445778247e-05	0.1	0.0004998917145070044	0.1	I(1)	0
cream	producer	daily	0.250564990586775	0.1	2.45386162864072e-07	0.1	5.803319568218489e-14	0.1	I(1)	0
	prozorro	daily	5.15934840442518e-22	0.1	3.6046356335359884e-13	0.1	1.1693001750830719e-11	0.09309066682789993	I(0)	0
cream	ultimate_consumer_price	daily	0.9898974984189568	0.04131706548274784	0.13281538679024207	0.1	1.8618636098860138e-07	0.1	I(2)	0
hard_cheese	producer	daily	0.07169849370926479	0.1	1.6796562524251144e-07	0.1	3.4022983690728456e-13	0.1	I(1)	0
hard_cheese	prozorro	daily	2.633366148124232e-10	0.08893253878307611	1.5324738060670524e-08	0.1	1.1965567303232429e-10	0.041666666666668085	I(0)	0
hard_cheese	ultimate_consumer_price	daily	0.0020093927286190185	0.1	1.0352889413033173e-06	0.1	3.4558512742049166e-06	0.1	I(0)	0
milk	producer	daily	0.11201722400789482	0.1	5.176893535234118e-05	0.1	2.027043992186184e-15	0.1	I(1)	0
	prozorro	daily	0.0014111002754477221	0.01	3.8189780115701814e-13	0.1	4.4125656034645e-11	0.1	I(1)	0
milk	ultimate_consumer_price	daily	0.03632042875304809	0.1	0.040537361406627226	0.1	2.2176533969954935e-05	0.1	I(0)	0
other	producer	daily	nan	nan	nan	nan	nan	nan	ambiguous	0
other	prozorro	daily	0.9029715543675231	0.01	1.3734510184087591e-14	0.1	1.260173359802745e-10	0.1	I(1)	1
other	ultimate_consumer_price	daily	0.5659999344514546	0.1	0.07783547487613032	0.1	0.001165657246792198	0.1	I(2)	0
sour_cream	producer	daily	0.1894666650765241	0.1	5.90273005994021e-14	0.1	4.493779268604255e-09	0.1	I(1)	0
sour_cream	prozorro	daily	0.21880956753864628	0.04947047925942609	2.411890357322739e-12	0.1	3.969186107838014e-13	0.1	I(1)	0
sour_cream	ultimate_consumer_price	daily	0.1170536345573358	0.1	5.055696436915593e-29	0.1	9.191766788720307e-17	0.1	I(1)	0

Table: Scaling_Factors

standardized_type	silpo_scale_to_consumer	novus_scale_to_consumer	n_overlap_silpo_consumer	n_overlap_novus_consumer	period_start	period_end
butter	1.0	1.0	0	0	2025-04-14	2026-01-08
cottage_cheese	1.0	1.0	0	0	2025-04-14	2026-01-08
cream	1.0	1.0	0	0	2025-04-15	2026-01-08
hard_cheese	0.9882560209867999	0.3947101267414928	38	32	2025-04-14	2026-01-08
milk	0.20159688129728354	0.15769761187472736	37	20	2025-04-14	2026-01-08
other	1.0	1.0	0	0	2025-04-14	2026-01-09
sour_cream	0.619069086264577	0.5746730019074469	38	12	2025-04-14	2026-01-08